Khalid Elbagir

golden-2024@hotmail.com | linkedin.com/in/khalid-elbagir | github.com/panther-0707 | Edinburgh, UK

Education

The University of Edinburgh — Edinburgh, UK
BEng (Hons) Computer Science — *Sept. 2024 – July 2028 (Expected)*

- **Relevant modules:** Intro to Algorithms and Data Structures, Software Engineering and Professional Practice, Computer Security, Software Testing, Discrete Mathematics and Probability, Reasoning and Agents

Experience

Edinburgh University Formula Student (EUFS) — Edinburgh, UK

Head of DevOps — ***Aug. 2026 – Present***

- Lead the sub-team that owns the team's developer platform: CLI tooling, GitLab CI/CD pipelines, and FastAPI/AWS S3 services
- Leading the Docker containerisation of the team's codebase and teaching teammates Linux, Docker and DevOps fundamentals

Software Engineering Intern, DevOps Sub-team — ***June 2026 – Aug. 2026***

- Audited three Python repositories (CLI tooling, FastAPI backends, ROS 2), producing a prioritised 39-issue backlog and fixing 19+ bugs, including a YAML deserialisation vulnerability enabling arbitrary code execution
- Built a Python tool ingesting vehicle telemetry packs from AWS S3 into a FastAPI/PostgreSQL platform via presigned-URL REST endpoints, streaming 150MB+ files with retry/backoff and cleanup of partial uploads
- Built a pytest suite from scratch for a 13-module CLI monorepo with zero prior coverage: 180+ unit tests using unittest.mock and filesystem fixtures, uncovering 4 latent production bugs
- Root-caused CI/CD failures in shared GitLab pipelines (dependency version mismatches, mypy module resolution collisions, rosdep misconfiguration) and authored an approved fix to shared team infrastructure

Software Team Member, DevOps Sub-team — ***Sept. 2025 – May 2026***

- Contributed bug fixes, features, and CI/CD and developer-tooling improvements across team repositories
- Contributed to the software behind EUFS's FSUK 2026 results: two FS-AI category wins (Ansys Autonomous Design; Simulation Development) and the first full ADS scrutineering pass by a UK team

Projects

gh-prompt-scan | *Python, Click, PyYAML, pytest, GitHub Actions* — June 2026

- Built and published an open-source static analysis tool detecting prompt injection vulnerabilities in AI-integrated GitHub Actions pipelines, based on 2026 security research from NUS and ByteDance
- Released on PyPI (`pip install gh-prompt-scan`) and the GitHub Actions Marketplace with configurable severity gating; pytest fixture pairs (vulnerable/safe) validate every threat vector

Samantha – AI Terminal Assistant | *Python, Docker, openEuler Linux, LLM APIs* — Oct. 2025

- Led a 5-person team to the final of the openEuler Challenge Week Hackathon, building an AI assistant that converts natural language into safe Linux shell commands using the Qwen3-Coder-480B LLM
- Engineered a natural-language-to-JSON planning pipeline with pronoun resolution ("find PDFs then move them"), fuzzy-matching typo self-correction, and content-aware file search, deployed via Docker.

Adventure-maps | *React, Python (FastAPI), MapLibre GL, OpenStreetMap, Vercel* — Oct. 2026

- Built a full-stack walking-route planner at a Hackathon that inverts navigation: instead of the fastest route, it spends a user's spare time on the most interesting route that still arrives on time
- Built a FastAPI backend with fallbacks across Overpass, OpenRouteService and OSRM to survive API outages.

Home Server – Self-Hosted Cloud | *Ubuntu Server, Docker, Nextcloud, Grafana, Pi-hole, Tailscale* — Oct. 2026

- Deployed a 24/7 self-hosted server on a repurposed 4 GB RAM laptop running six Docker containers: Nextcloud (app and DB), Prometheus, Grafana, Uptime Kuma, Pi-hole) at no hardware or software cost
- Secured remote access with Tailscale (WireGuard VPN), a UFW firewall and automatic updates, with no router ports opened; monitored with Prometheus, Grafana and 7 Uptime Kuma checks

Technical Skills

Languages: Python, Java, C, SQL, JavaScript, Bash, HTML/CSS, C++
Frameworks & Libraries: FastAPI, SQLAlchemy, Pydantic, Spring Boot, pandas, ROS 2, REST APIs
Testing & Code Quality: pytest, unittest.mock, pytest-cov, flake8, mypy, static analysis
Developer Tools: Git, GitLab CI/CD, GitHub Actions, Docker, Docker Compose, Linux, Tailscale, Prometheus, Grafana, AWS S3 (boto3), PostgreSQL

Leadership & Achievements

Class Representative, The University of Edinburgh (Sept. 2025 – May 2026): represented 400+ students
Awards: Duke of Edinburgh Silver Award; Brown Belt in Karate
Certifications: First Aid (British Red Cross); SCQF Certificates: Communication, Conflict Resolution